César Antonio Carvajal de la Haza

Mathematician · Statistician · Physicist

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QUALIFICATION SUMMARY

Triple graduate in Mathematics, Statistics, and Physics with a strong foundation in quantum mechanics, theoretical physics, statistical methods, and applied mathematics. Experienced in Python, R, and mathematical modelling, with hands-on projects in GNSS/IMU sensor fusion, quantum entanglement entropy, and data analysis. Passionate about quantum computing, artificial intelligence, and space science. Currently expanding expertise in quantum computing and Linux/C++ systems programming.

EDUCATION

Bachelor's Degree in Physics — 240 ECTS (2021 – 2025) | GPA: 8.1/10

National University of Distance Education (UNED), Spain

- Strong background in classical and quantum physics, optics, and dynamical systems.
- Fields of interest: Quantum Physics, General Relativity, Theoretical Physics, Nuclear Physics, Optics.
- Honours in Classical Mechanics, Vibrations & Waves, Quantum Physics II, and Dynamical Systems.
- Final Grade Project: *Quantum Mechanics and the study of Entanglement Entropy and Geometry* — implemented Python simulations to analyse entanglement in quantum systems.

Bachelor's Degree in Mathematics — 240 ECTS (2016 – 2021) | GPA: 8.0/10

University of Seville, Spain

- Strong foundation in calculus, algebra, and differential equations.
- Fields of interest: Computing, Algebra, Ordinary & Partial Differential Equations, Topology, Probability, Coding Theory & Cryptography.
- Gained experience in Haskell, Python, Maxima, and MATLAB; developed solid programming fundamentals through mathematical problem-solving.
- Particular interest in Algebra and the structures formed by number systems.

Bachelor's Degree in Statistics — 240 ECTS (2016 – 2021) | GPA: 8.3/10

University of Seville, Spain

- Expertise in probability, statistical inference, and data analysis.
- Fields of interest: Statistical Inference, Databases, Survey Design, Operations Research, Multivariate Analysis, Computational Statistics, Decision Theory, AI & Statistics.
- Developed multiple projects in R and Excel; introduced to Machine Learning algorithms throughout the programme.
- Honours in Survey Design; participated in a real-world user-satisfaction study at the University of Seville.

TRAINING & CONTINUING EDUCATION

Introduction to Quantum Computing — 1.5 ECTS (2026 – ongoing)

UNED, Spain · Extension course

- Foundational concepts in quantum computing: qubits, quantum gates, superposition, entanglement, and quantum algorithms.

GNSS Academy — Intensive GNSS Training Programme (January 2025 – July 2025)

GNSS Academy, Spain · 6-month combined study and practical programme

- Comprehensive study of Global Navigation Satellite System (GNSS) theory, architectures, and signals.
- Topics covered: DGNSS, RTK, PPP, SBAS, GNSS data processing, ionospheric and tropospheric corrections, multi-path mitigation, satellite orbit propagation (SP3), and advanced positioning techniques.
- Hands-on use of gLAB for analysing GNSS observables and signals.
- **Capstone project:** Full-featured GNSS+IMU sensor fusion simulator developed in Python/Linux:
  - Simulated GNSS measurements using precise SP3 orbit propagation and realistic error models (ionospheric, tropospheric, clock, multipath).

- Applied Antenna Phase Offset (APO) and Sagnac effect corrections.
- Implemented mechanisation equations in ECEF frame to derive position, velocity, and attitude from simulated IMU data.
- Performed coordinate transformations for accurate motion tracking.
- Implemented a Least-Squares Estimation module for GNSS-only positioning.

WORK EXPERIENCE

Secondary School Teacher — Mathematics (2026 – present)

Instituto de Educación Secundaria, Spain

- Teaching Mathematics from 3º ESO through 2º Bachillerato, covering algebra, calculus, statistics, and geometry.
- Preparation of exams, tests, and learning materials adapted to each educational level.
- Curriculum planning and organisation; design of personalised adaptations for students with diverse learning needs.
- Active participation in departmental meetings and academic committees.
- Providing academic guidance and study-skills coaching to students.

QA Automation Engineer (April 2022 – November 2022)

Deloitte, Spain

- Developed automated test suites using Selenium and Java, significantly reducing manual testing time.
- Co-implemented an automation framework using Selenium, Cucumber, and Jenkins, improving coverage and efficiency.
- Executed test scripts, documented results, and created detailed bug reports.
- Identified and prioritised defects using Jira; maintained synchronised codebase with Git.
- Collaborated with cross-functional teams in an international environment.

Private Tutor — Mathematics, Physics, Chemistry & Technical Drawing (several years)

Self-employed

- Tutored students across multiple subjects, improving their academic performance and confidence.
- Created a supportive environment focused on clarity, logical reasoning, and problem-solving.

RESEARCH & ACADEMIC / PERSONAL PROJECTS

MLWeather — Temperature Forecasting with ML (2026)

Personal project · GitHub · Live app

- End-to-end pipeline for temperature forecasting in Seville using hourly meteorological data from Open-Meteo API.
- Compared 5 models: Linear Regression, XGBoost, Random Forest, SARIMA, and LSTM; evaluated with MAE, RMSE, and R^2.
- Applied preprocessing techniques including lag features, cyclical time encodings (sin/cos), and strict train/test separation to avoid data leakage.
- Built an interactive Streamlit app with modular architecture (src/, models/, ui/, evaluation/) for exploring and comparing model predictions.

Learning Linux, C/C++ and Bash (ongoing)

Self-directed

- Building foundational skills in Linux, Bash scripting, file handling, and C/C++ programming.
- Practising automation and scripting through small hands-on exercises.

Entanglement Entropy in Quantum Systems (2024 – 2025)

Final degree project — Physics

- Studied entanglement entropy and its dependence on system geometry to test the limits of the Area Law.
- Implemented numerical simulations in Python. Code available on GitHub.

Survey Design & Statistical Analysis (2019/20)

University of Seville — Statistics degree

- Participated in the 2019/20 User Satisfaction Study of the IT Service at the University of Seville.
- Contributed to survey design, data collection, statistical analysis in R and Excel, and final report preparation.

Problem-Solving & Coding Challenges (2016 – 2018)

Self-directed

- Solved algorithmic problems on Exercitium and Kattis to develop problem-solving and computational thinking skills.

TECHNICAL SKILLS

Programming Languages: Python, R, Haskell, MATLAB, SQL (basic), C/C++ (basic), Bash (basic)

Tools & Platforms: Linux, Git, VSCode, Jupyter, LaTeX, gLAB

GNSS & Navigation: DGNSS, RTK, PPP, SBAS, GNSS data processing, SP3 orbit propagation, IMU sensor fusion, ECEF transformations

Data Analysis & Modelling: Applied statistics, mathematical modelling, machine learning, optimisation

Applied Mathematics: Algebra, differential equations, cryptography, topology, optimisation

Quantum: Quantum mechanics, entanglement entropy, quantum computing (introductory)

SOFT SKILLS

Critical Thinking · Analytical Skills · Creative Thinking · Attention to Detail · Adaptability · Research Skills · Interpersonal Skills

LANGUAGES

Spanish (Native) · English (C1 — Proficient) · French (Basic, in progress)

EXTRACURRICULAR ACTIVITIES & VOLUNTEERING

LGTB+ Advocacy & Awareness (2017 – 2019)

Colectivo LGTB+ Pedro Zerolo, Faculty of Mathematics, University of Seville

- Participated in awareness campaigns and outreach activities related to the LGBT+ community.

Animal Welfare Volunteering

Local animal shelter

- Provided weekly care for rescued dogs; created adoption advertisements and participated in awareness campaigns.

INTERESTS

Quantum Physics · Artificial Intelligence · Data Visualisation · Language Learning · Writing · Crossfit